MULTIPLE COMPARISON TESTS

TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 19

5-MINUTE REVISION

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1. QUICK REVISION · 1

- ANOVA → omnibus test for detecting an overall treatment effect.
- Significant ANOVA → indicates that at least one pair of treatment means differs.
- ANOVA alone → does not identify which specific treatment pairs differ.
- Multiple comparison tests → used to identify specific differences among treatment means.
- Protected Fisher's LSD requires a significant ANOVA; Tukey's HSD does not require this preliminary test.

2. QUICK REVISION · 2

- Type I error → rejecting a true null hypothesis.
- Type II error → failing to reject a false null hypothesis.
- Family → complete set of pairwise comparisons in an experiment.
- FWER → probability of making at least one Type I error within the family of comparisons.
- Number of pairwise comparisons → $v(v-1)/2$.

3. QUICK REVISION · 3

- Fisher's LSD → two-step procedure involving significant ANOVA followed by pairwise comparisons.
- LSD → Least Significant Difference.
- CD → Critical Difference; commonly used term for LSD in Indian agricultural research.
- Protected LSD → pairwise comparisons are performed only after a significant ANOVA.
- Decision under LSD → if $|\overline{T_i} - \overline{T_j}| > \text{LSD}$, the means differ significantly.

4. QUICK REVISION · 4

- Equal replications → $\text{LSD} = t_{\alpha/2,edf}\sqrt{2\text{MSE}/r}$.
- LSD is relatively liberal compared with Tukey's test.
- Fisher's LSD controls experiment-wise Type I error effectively for two or three treatments under the usual model, but not generally for more than three, as discussed in the chapter.
- DMRT → Duncan's Multiple Range Test.
- DMRT → useful when comparing many treatment means and widely used in plant sciences.

5. QUICK REVISION · 5

- DMRT procedure → rank means, calculate standard error, obtain r_p, calculate R_p, and compare differences with the appropriate R_p.
- In DMRT, p → number of ordered treatment means spanned by the comparison, including the two means being compared.
- Adjacent means → $p = 2$.
- One mean between the two means → $p = 3$.
- Two means between the two means → $p = 4$.

6. QUICK REVISION · 6

- DMRT critical range → increases as p increases.
- DMRT → generally more powerful and more liberal than Tukey's test.
- Tukey's HSD → compares all possible pairs while controlling the family-wise error rate.
- Tukey's test → based on the studentised range statistic q.
- HSD → single critical difference for equal replications.

7. QUICK REVISION · 7

- Decision under Tukey → if $|\overline{T_i} - \overline{T_j}| > \text{HSD}$, the means differ significantly.
- Tukey's test → more conservative than LSD and DMRT.
- More conservative test → less likely to declare a difference significant.
- In the worked example → LSD = 1.10.
- In the worked example → DMRT R_p values range from 1.10 to 1.21.

8. QUICK REVISION · 8

- In the worked example → Tukey HSD = 1.59.
- Under LSD and DMRT → all treatment pairs are significant.
- Under Tukey → differences of 1.5 are not significant.
- Tukey therefore identifies fewer significant differences than LSD and DMRT in the example.
- Compact letter grouping → convenient presentation of multiple comparison results.

9. QUICK REVISION · 9

- Same letter → treatments are not significantly different.
- No common letter → treatments are significantly different.
- A treatment can have more than one letter when it belongs to overlapping non-significant groups.
- Letter grouping is also called alphabet notation.
- Line or underscore notation → older method of presenting groups of means.

10. QUICK REVISION · 10

- Letter and line notation → convey the same statistical information.
- Dunnett's test → useful when every treatment is compared with a single control.
- Bonferroni method → adjusts significance for multiple comparisons.
- Scheffé's test → useful for general contrasts among means.
- SNK → Student-Newman-Keuls multiple comparison procedure.

11. IMPORTANT FORMULAE · 1

Number of pairwise comparisons among v treatments:

$$N = \binom{v}{2} = \frac{v(v-1)}{2}$$

Inflated Type I error rate:

$$\text{Inflated } \alpha = 1 - (1-\alpha)^N$$

Standard error of the difference between two treatment means:

$$SE(d) = \sqrt{\text{MSE}\left(\frac{1}{r_i} + \frac{1}{r_j}\right)}$$

12. IMPORTANT FORMULAE · 2

Fisher's LSD:

$$\text{LSD} = t_{\frac{\alpha}{2}, edf} \cdot SE(d)$$

LSD for equal replications:

$$\text{LSD} = t_{\frac{\alpha}{2}, edf} \cdot \sqrt{\frac{2\text{MSE}}{r}}$$

Test statistic for pairwise comparison:

$$t = \frac{\overline{T_i} - \overline{T_j}}{SE(d)}$$

13. IMPORTANT FORMULAE · 3

Standard error of a treatment mean in DMRT:

$$SE(\overline{Y}) = \sqrt{\frac{\text{MSE}}{r}}$$

Shortest significant range in DMRT:

$$R_p = r_p \cdot SE(\overline{Y})$$

Tukey's honestly significant difference:

$$\text{HSD} = q_{\alpha,(v,edf)} \cdot \sqrt{\frac{\text{MSE}}{r}}$$

14. SELF-CHECK FROM THE CHAPTER

1. A test of the overall equality of several treatment means is called an _____ test. **[Omnibus]**
2. A significant treatment F-test indicates that not all treatment means are _____. **[Equal]**
3. The probability of at least one Type I error in a family of tests is called the _____ error rate. **[Family-wise]**
4. The number of pairwise comparisons among v treatment means is _____. **[$v(v-1)/2$]**